Complete example, new exception class definition in python, only code no words

python

```python
class CustomValidationError(Exception):
    def __init__(self, message, field=None, code=None):
        self.message = message
        self.field = field
        self.code = code
        super().__init__(self.message)

    def __str__(self):
        parts = [self.message]
        if self.field:
            parts.append(f"field={self.field}")
        if self.code:
            parts.append(f"code={self.code}")
        return " | ".join(parts)


class User:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def validate(self):
        if not self.name:
            raise CustomValidationError("Name cannot be empty", field="name", code="E0
01")
        if not isinstance(self.age, int):
            raise CustomValidationError("Age must be an integer", field="age", code="E
002")
        if self.age < 0 or self.age > 150:
            raise CustomValidationError("Age out of range", field="age", code="E003")


def main():
    users = [
        User("Alice", 30),
        User("", 25),
        User("Bob", "thirty"),
        User("Charlie", 200),
    ]
```

```
    for u in users:
        try:
            u.validate()
            print(f"OK: {u.name}, {u.age}")
        except CustomValidationError as e:
            print(f"ValidationError: {e}")
        except Exception as e:
            print(f"Unexpected: {type(e).__name__}: {e}")


if __name__ == "__main__":
    main()
```

How to join a list of stirngs

python

```python
# Join a list of strings with a separator
words = ["hello", "world", "python"]

result = " ".join(words)
print(result)  # hello world python

result = ", ".join(words)
print(result)  # hello, world, python

result = "".join(words)
print(result)  # helloworldpython

# Join with newline
result = "\n".join(words)
print(result)
# hello
# world
# python

# Join non-string items (must convert first)
numbers = [1, 2, 3]
result = "-".join(str(n) for n in numbers)
print(result)  # 1-2-3

# Or using map
result = "-".join(map(str, numbers))
print(result)  # 1-2-3
```